

My Project report

My dataset is Titanic dataset it was good dataset for learning ML . Also it is easier than other datasets options because it takes shorter time to clean data . Because of that I choose Titanic dataset .

The dataset has 891 rows and 12 columns . Classes are balanced(0: 549, 1 : 342) . I fill the missing values in Age columns . And I dropped some columns that is not important in prediction surviving . I did not drop any rows . I make label encoding in the Sex column because I cannot use it without encoding . I split the data into X train and test (7 columns) and y train and test

I trained three models: Logistic Regression ($F1 = 0.72$), Random Forest ($F1 = 0.71$), and KNN ($F1 = 0.32$). Logistic Regression was best. Choosing the best was by $F1$, because I want both recall and precision . And $F1$ combines between them.

My network had one Dense layer (128 neurons) with Dropout (the rate is 0.3).I took 10% of the train set for validation. After 10 epochs validation accuracy reached 70%. The optimizer is Adam. And the metrics I used are (precision , recall, accuracy)

My neural network has the lowest accuracy (0.64) . The other models(like Logistic Regression) have higher accuracy and take shorter time than the neural network . so it is better to use the other models in this project because it is faster and give us better accuracy .

Lastly , I learned after I finished my project how to make neural network on binary classification . I learned that numbers of layers have a relationship with the overfitting .and I learned that reducing layers reduce overfitting . Also , my skill in making models became better than before .

. After cleaning , handling, and preparing the data became 7 columns in X_train and X_test and one column in y_train and Y_test . The columns that were dropped are not important in prediction so, I dropped it. I didn't drop any rows . I make one-hot encoding to the Sex column .